

## 1) Week 11: Numerical differentiation

### 1) Week 11: Numerical differentiation

1.1) The distance  $x$  of a runner from a fixed point is measured in meters at intervals of half a second. The data obtained are:

| $t$ | $x$   |
|-----|-------|
| 0.0 | 0.0   |
| 0.5 | 3.65  |
| 1.0 | 6.80  |
| 1.5 | 9.90  |
| 2.0 | 12.15 |

Use forward, backward, and central differences to approximate the runner's velocity at times  $t = 1.0$  s.

#### • Forward

$$x'_F(1.0) = \frac{x(1.5) - x(1.0)}{h} = \frac{9.9 - 6.8}{0.5} = 6.2$$

#### • Backward

$$x'_B(1.0) = \frac{x(1.0) - x(0.5)}{h} = \frac{6.8 - 3.65}{0.5} = 6.3$$

#### • Central

$$x'_C(1.0) = \frac{x(1.5) - x(0.5)}{2h} = \frac{9.9 - 3.65}{1.0} = 6.25$$

1.2) Suppose that you take samples of sea level heights at a rate of once every two hours, and you use the backward differences scheme to estimate the rate of change.

- a) Compute a bound on the size of the error incurred if sea level is sinusoidal with an amplitude of 3 feet and a period of 12 hours, meaning that it is generated by the function  $f(t) = 3 \sin(\frac{2\pi}{12}t)$ , with  $t$  measured in hours.
- b) How often should you take measurements if you wish the backward differencing error to not exceed 0.1 ft/hr?
- c) Repeat using the central differencing scheme.

$$a) |E_B(h)| \leq \frac{M_2}{2} h$$

$$\text{where } M_2 = \max_{t \in \mathbb{R}} |f''(t)|$$

$$f(t) = A \sin \omega t$$

$$f'(t) = A \omega \cos \omega t$$

$$f''(t) = -A \omega^2 \sin \omega t$$

$$\therefore M_2 = \max_{t \in \mathbb{R}} |-A \omega^2 \sin \omega t| = A \omega^2$$

$$\therefore |E_B(h)| \leq \frac{A \omega^2}{2} h = \frac{3 \left(\frac{2\pi}{12}\right)^2}{2} 2 = 0.822 \frac{\text{ft}}{\text{hr}}$$

$$b) \text{ Want } h \text{ such that } \frac{1}{2} A \omega^2 h < 0.1$$

$$\therefore \text{ need } h < \frac{2 \cdot 0.1}{A \omega^2} = \frac{2 \cdot 0.1}{3 \left(\frac{2\pi}{12}\right)^2} = 0.243 \text{ hr}$$

c) • Error bound with  $h=2$ .

$$f'''(t) = -A\omega^3 \cos \omega t \rightarrow M_3 = A\omega^3$$

$$|E_c| \leq \frac{M_3}{6} h^2 = \frac{A\omega^3}{6} h^2 = \frac{1}{6} 3 \left(\frac{2\pi}{12}\right)^3 \cdot 4 = 0.287 \frac{\text{ft}}{\text{hr}}$$

•  $h$  for error bound  $\leq 0.1$

$$\frac{1}{6} 3 \left(\frac{2\pi}{12}\right)^3 \cdot h^2 \leq 0.1$$

$$\therefore h \leq \sqrt{\frac{0.2}{(2\pi/12)^3}} = 1.18 \text{ hr}$$

## 2) Week 11: Numerical integration

2.1) Suppose we have taken the following samples of a function  $f(x)$ .

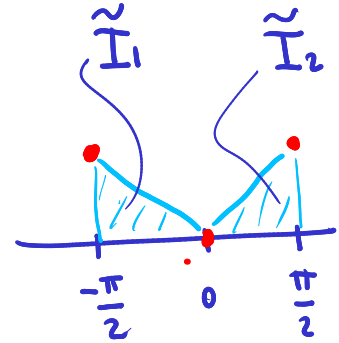
| $x$              | $f(x)$ |
|------------------|--------|
| $-\frac{\pi}{2}$ | 1      |
| 0                | 0      |
| $\frac{\pi}{2}$  | 1      |

Use this data to estimate

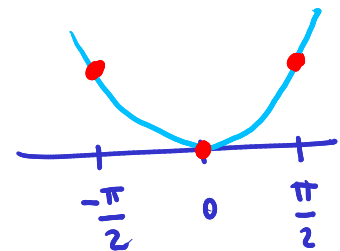
$$\int_{-\pi/2}^{\pi/2} f(x) dx$$

- a) with the composite trapezoidal rule
- b) with the composite Simpson's 1/3 rule

$$\begin{aligned} \text{a) } \tilde{I} &= \tilde{I}_1 + \tilde{I}_2 \\ &= \frac{h}{2} (f_1 + f_2) + \frac{h}{2} (f_2 + f_3) \\ &= \frac{\pi/2}{2} (1 + 0 + 0 + 1) = \frac{\pi}{2} \end{aligned}$$

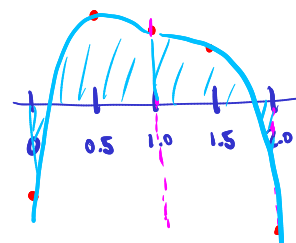


$$\begin{aligned} \text{b) } \tilde{I} &= \frac{h}{3} (f_0 + 4f_1 + f_2) \\ &= \frac{\pi}{6} (1 + 0 + 1) = \frac{\pi}{3} \end{aligned}$$



**2.2)** Use the following table of samples of  $f(x)$  to approximate  $\int_0^{2.5} f(x)dx$  using composite Simpson's 1/3 rule.

| $x$ | $f(x)$ |
|-----|--------|
| 0.0 | -2.5   |
| 0.5 | 3.0    |
| 1.0 | 2.5    |
| 1.5 | 2.0    |
| 2.0 | -4.5   |



$$\tilde{I} = \tilde{I}_1 + \tilde{I}_2$$

$$= \frac{h}{3} (f_0 + 4f_1 + f_2) + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

$$= \frac{0.5}{3} (-2.5 + 4 \cdot 3 + 2.5) + \frac{0.5}{3} (2.5 + 4 \cdot 2.0 - 4.5)$$

$$= 3$$

2.3) We wish to approximate the integral:

$$\int_0^3 x^4 dx$$

Find a bound on the approximation error of the following two techniques:

- a) Composite trapezoidal with  $h = 0.2$ .
- b) Composite Simpson's 1/3 rule with  $h = 0.1$ .

$$f(x) = x^4$$

$$f'(x) = 4x^3$$

$$f''(x) = 12x^2 \longrightarrow M_2 = 12 \cdot 9 = 108$$

$$a) \text{ error} \leq \frac{(b-a)h^2}{12} M_2 = \frac{3 \cdot 0.2^2}{12} \cdot 108 = 1.08$$

$$b) \quad f'''(x) = 24x$$

$$f''(x) = 24 \longrightarrow M_4 = 24$$

$$\text{error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{3 \cdot (0.1)^4}{180} 24 = 4 \cdot 10^{-5}$$

2.4) We wish to approximate the integral of  $f(x) = cx^2 + 3x + 4$  over the interval  $[-6, 6]$  with a step size of  $h = 0.5$ . Find the maximum value of  $c$  such that the error is guaranteed not to exceed 1 when the approximation is made with

- a) the composite trapezoidal rule, or
- b) the composite Simpson's rule.

a)  $f'(x) = 2cx + 3$

$$f''(x) = 2c \longrightarrow M_2 = 2|c|$$

$$\begin{aligned} \text{error} &\leq \frac{(b-a)h^2}{12} M_2 = \frac{(6+6)h^2}{12} \cdot 2|c| \\ &= 2h^2|c| \end{aligned}$$

Want this to be less than 1.

$$2h^2|c| < 1 \iff |c| < \frac{1}{2h^2} = \frac{1}{2 \cdot \frac{1}{4}} = 2$$

$\therefore$  maximum  $C = 2$ .

b)  $f'''(x) = 0$

$$f^{(4)}(x) = 0 \longrightarrow M_4 = 0$$

$\therefore$  error  $= 0$  for any value of  $c$ .

2.5) We wish to approximate the integral of  $f(x) = e^{-x} \sin(x)$  on the interval  $[0, 3]$  using a segment size of  $h = 0.1$ . Find an upper bound on the error made by

a) the composite trapezoidal rule

b) the composite Simpson's rule. (Note: Your estimate of  $M_4$  does not have to be exact).

$$a) \quad f'(x) = -e^{-x} \sin x + e^{-x} \cos x = e^{-x} (\cos x - \sin x)$$

$$\begin{aligned} f''(x) &= -e^{-x} (\cos x - \sin x) + e^{-x} (-\sin x - \cos x) \\ &= -2e^{-x} \cos x \end{aligned}$$

$$M_2 = \max_{x \in [0, 3]} |f''(x)| = \max_{[0, 3]} 2e^{-x} \cos x = 2$$

$$\therefore \text{error} \leq \frac{(b-a)h^2}{12} M_2 = \frac{3(0.1)^2}{12} \cdot 2 = 0.005$$

$$b) \quad f'''(x) = 2e^{-x} \cos x + 2e^{-x} \sin x = 2e^{-x} (\sin x + \cos x)$$

$$\begin{aligned} f^{(4)}(x) &= -2e^{-x} (\sin x + \cos x) + 2e^{-x} (\cos x - \sin x) \\ &= -4e^{-x} \sin x \end{aligned}$$

$$M_4 = \max_{[0, 3]} 4e^{-x} |\sin x| \leq 4 \quad \dots \text{this is a conservative estimate, but it's the best I can do by hand.}$$

$$\therefore \text{error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{3(0.1)^4}{180} \cdot 4 = 6.6 \cdot 10^{-6}$$



2.6) For which of the following functions  $f(x)$  is the estimate of

$$\int_0^{10} f(x) dx$$

using composite Simpson's rule with  $h = 1$  guaranteed to produce an error less or equal to 0.5?

a)  $f_1(t) = 3 \sin(t)$

b)  $f_2(t) = e^{t/10}$

c)  $f_3(t) = t^5$

d)  $f_4(t) = 2t$

$$\text{In general, error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{10 \cdot 1}{180} M_4 < 0.5$$

$$\text{Error} \leq 0.5 \text{ when } M_4 \leq \frac{180}{10} \cdot 0.5 = 9$$

a)  $f'_1 = 3 \cos t$

$$f''_1 = -3 \sin t$$

$$f'''_1 = -3 \cos t$$

$$f^{(4)}_1 = 3 \sin t \rightarrow M_4 = 3 \quad \checkmark$$

b)  $f'_2 = \frac{1}{10} e^{t/10}$

$$f''_2 = \frac{1}{100} e^{t/10}$$

$$f'''_2 = \frac{1}{1000} e^{t/10}$$

$$f^{(4)}_2 = \frac{1}{10000} e^{t/10} \rightarrow M_4 = \max_{(0,10)} \frac{e^{t/10}}{10000} = \frac{e}{10,000} < 9 \quad \checkmark$$

c)  $f'_3 = 5 t^4$

$$f''_3 = 20 t^3$$

$$f'''_3 = 60 t^2$$

$$f^{(4)}_3 = 120 t$$

$$\rightarrow M_4 = \max_{[0,10]} 120 t = 1200 > 9 \quad \times$$

d)  $f^{(4)}_4 = 0 \rightarrow M_4 = 0 \rightarrow \text{error} = 0 \quad \checkmark$

$$\begin{aligned}
 S_m = f(y_m) &= f((0.25, 1.25)) \\
 &= (1.25, (1-0.25)^2(1.25) - 0.25) \\
 &= (1.25, 0.453125) = \hat{S}
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}[1] &= \hat{y}[0] + 0.5 \hat{S} \\
 &= (0, 1) + 0.5 (1.25, 0.453125) \\
 &= (0.625, 1.23)
 \end{aligned}$$

### 3) Week 12: Bisection and Newton-Raphson

3.1) Under what conditions might the bisection method fail to find a root of  $f(x)$ ? What about Newton-Raphson?

- Bisection may fail if:
  - >  $f(x)$  has no roots.
  - >  $f(x)$  is not continuous
- Newton Raphson may fail if
  - >  $f(x)$  has no roots
  - >  $f(x)$  is not differentiable
  - >  $f(x)$  has locations where  $f'(x)=0$

3.2) By what factor does the interval that contains the root decrease in each iteration of the bisection method?

By a factor of 2

3.3) Carry out the first three iterations of the Newton-Raphson method for the function  $f(x) = x^2 - 4$  with initial value  $x_0 = 6$ .

$$f'(x) = 2x$$

$$\text{Newton Raphson: } x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$= x_k - \frac{x_k^2 - 4}{2x_k}$$

$$= \frac{x_k}{2} + \frac{2}{x_k} \quad \frac{68}{30} = \frac{34}{15}$$

$$x_1 = \frac{x_0}{2} + \frac{2}{x_0} = 3 + \frac{1}{3} = \frac{10}{3}$$

$$x_2 = \frac{x_1}{2} + \frac{2}{x_1} = \frac{10}{6} + \frac{6}{10} = \frac{100 + 36}{60} = \frac{136}{60} = \frac{34}{15}$$

$$x_3 = \frac{136}{120} + \frac{120}{136} = 2.02.$$

3.4) How many iterations will be taken by the bisection method to find a root of the function  $f(x) = x^2 - 11$  if the initial interval is  $[0, 5]$  and the tolerance is 0.5?

$$K_{\min} = \text{ceil } \log_2 \frac{5-0}{0.5} = 4$$

3.5) You wish to use the bisection method to find a root of  $f(x) = x^2 - 3x + 1$  to within a tolerance of  $10^{-3}$ .

- Suggest a viable initial condition for the iteration.
- Compute the interval after two iterations.
- How many iterations will it take for the algorithm to converge given your suggested initial condition?

a) Need  $N_0$  and  $P_0$  such that  $f(N_0) < 0$  and  $f(P_0) > 0$ .

$$P_0 = 0 \rightarrow f(P_0) = 1 \quad \checkmark$$

$$N_0 = 1 \rightarrow f(N_0) = 1 - 3 + 1 = -1 \quad \checkmark$$

b)

| $k$ | $N_k$ | $P_k$ | $C$   | $f(C)$                           |
|-----|-------|-------|-------|----------------------------------|
| 0   | 1     | 0     | 0.5   | $0.25 - 1.5 + 1 = -0.25 < 0$     |
| 1   | 0.5   | 0     | 0.25  | $0.25^2 - 0.75 + 1 = 0.3125 > 0$ |
| 2   | 0.5   | 0.25  | 0.375 |                                  |

c)

$$k_{\min} = \text{ceil} \log_2 \frac{1-0}{10^{-3}} = 10$$

3.6) You wish to use the bisection method to find a root of  $f(x) = e^{-x} \cos(x)$  in the interval  $[0, 3]$ .

a) Is  $[0, 3]$  a viable initial interval for the iteration? If not, suggest another.

b) Create a table with the first three iterations of the method.

c) How many iterations are needed to find a root to within a tolerance of  $10^{-5}$ ?

$$\begin{aligned} a) \quad f(0) &= e^{-0} \cos 0 = 1 > 0 \\ f(3) &= e^{-3} \cos 3 = -0.05 < 0 \end{aligned}$$

We have a negative and a positive evaluation, so yes.

b)

| $k$ | $N_k$ | $P_k$ | $C$    | $f(C)$        |
|-----|-------|-------|--------|---------------|
| 0   | 3     | 0     | 1.5    | $0.1578 > 0$  |
| 1   | 3     | 1.5   | 2.25   | $-0.0662 < 0$ |
| 2   | 2.25  | 1.5   | 1.875  | $-0.046 < 0$  |
| 3   | 1.875 | 1.5   | 1.6875 | $-0.021$      |

$$c) \quad k_{\min} = \text{ceil} \log_2 \frac{3}{10^{-5}} = 19$$

3.7) You wish to find a root of the function  $f(x) = e^{-x} - x$ , defined on the real line.

- Will Newton-Raphson converge to the root regardless of the initial condition? Why or why not?
- Carry out the first two iterations of Newton-Raphson, starting from  $x_0 = 1$ .

a) We need  $f(x)$  to be continuous and never have  $f'(x) = 0$ .

- $f(x)$  is continuous everywhere ✓

- $f'(x) = -e^{-x} - 1 = 0 \iff e^{-x} = -1 \dots$  impossible.

$\therefore$  We expect N.R. to converge to a root, if one exists, for any  $x_0$ .

$$\begin{aligned} \text{b) } x_{k+1} &= x_k - \frac{f(x_k)}{f'(x_k)} = x_k + \frac{e^{-x_k} - x_k}{e^{-x_k} + 1} \\ \therefore &= \frac{x_k e^{-x_k} + \cancel{x_k} + e^{-x_k} - \cancel{x_k}}{e^{-x_k} + 1} \\ &= \frac{x_k + 1}{e^{x_k} + 1} \end{aligned}$$

$$x_1 = \frac{2}{e+1} = 0.5379$$

$$x_2 = \frac{0.5379 + 1}{e^{0.5379} + 1} = 0.5669$$

3.8) You wish to find a root of the function  $f(x) = x^3 - 3x + 1$  using the Newton-Raphson method. Give an initial condition for which the approach is guaranteed to fail.

It will fail if  $f'(x_0) = 0$ .

$$f'(x) = 3x^2 - 3$$

$$f'(x) = 0 \Leftrightarrow 3x^2 - 3 = 0$$

$$\Leftrightarrow x^2 - 1 = 0$$

$$\Leftrightarrow x = \pm 1$$



3.9) You wish to use the bisection method to find a root of the function:

$$f(x) = (x - 2.5)(x + 1)(x - 1)$$

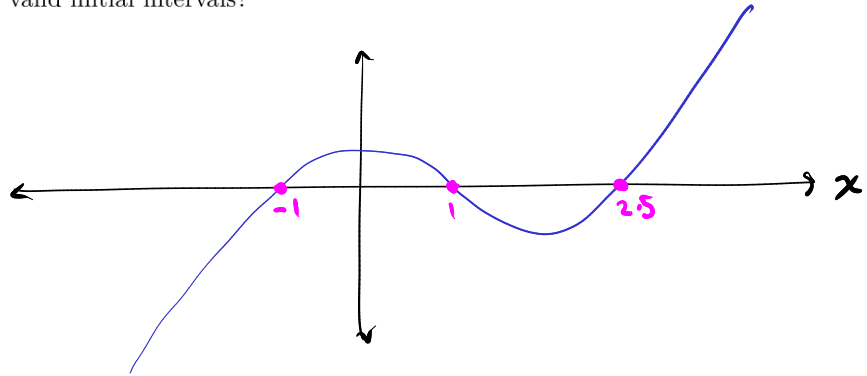
Which of the following options are valid initial intervals?

☐  $[-1, 1]$

☐  $[0, 1]$

☐  $[0, 2]$

☐  $[0, 3]$



For an initial interval  $[a, b]$  to be valid we need  
either  $f(a) > 0$ ,  $f(b) < 0$  or  $f(b) < 0$ ,  $f(a) > 0$ .

•  $[-1, 1]$  :  $f(a) = f(b) = 0$  X

•  $[0, 1]$  :  $f(0) > 0$ ,  $f(1) = 0$  X

•  $[0, 2]$  :  $f(0) > 0$ ,  $f(2) < 0$  ✓

•  $[0, 3]$  :  $f(0) > 0$ ,  $f(3) > 0$  X

#### 4) Week 12: Fixed points

4.1) We wish to find a root of the function

$$f(x) = x^2 - \cos x$$

a) Which of the following functions  $g$  have a fixed point that is a root of  $f$ ?

☐  $g_1(x) = x + x^2 - \cos x$

☐  $g_2(x) = \sqrt{\cos x}$

☐  $g_3(x) = x - \cos x$

•  $x$  is fixed point of  $g_1 \Rightarrow \cancel{x + x^2 - \cos x} = \cancel{x}$

$\Rightarrow x^2 - \cos x = 0 \Rightarrow f(x) = 0$

✓

•  $x$  is fixed point of  $g_2 \Rightarrow \sqrt{\cos x} = x$

$\Rightarrow \cos x = x^2 \Rightarrow f(x) = 0$

✓

•  $x$  is fixed point of  $g_3 \Rightarrow x - \cos x = x$

$\Rightarrow \cos x = 0 \not\Rightarrow f(x) = 0$

✗

For example  $x = \frac{\pi}{2}$  is f.p. of  $g_3$  but not root of  $f$ .

b) Suppose that we know that a root exists in the interval  $[0.5, 1]$ . For which functions  $g$  is the fixed point iteration guaranteed to converge when initialized in that interval? (the plot of  $\frac{\sin x}{\sqrt{\cos x}}$  is shown for reference).

☐  $g_1(x) = x + x^2 - \cos x$

☐  $g_2(x) = \sqrt{\cos x}$

☐  $g_3(x) = x - \cos x$

•  $g_1: \quad g'_1(x) = 1 + 2x + \sin x$

$$\max_{[0.5, 1]} |g'_1(x)| = \max_{[0.5, 1]} \underbrace{1 + 2x}_{\text{positive}} + \underbrace{\sin x}_{\text{positive}} > 1$$

✗

This is not guaranteed to converge.

•  $g_2: \quad g'_2(x) = -\frac{1}{2} \frac{\sin x}{\sqrt{\cos x}}$

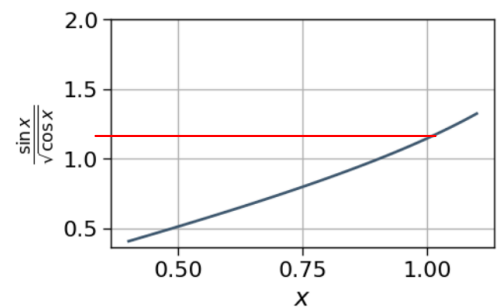
For  $\max_{[0.5, 1]} |g'_2(x)| < 1$  to

hold, we need

$$\left| -\frac{1}{2} \frac{\sin x}{\sqrt{\cos x}} \right| < 1 \quad \forall x \in [0.5, 1]$$

Or, what is the same,

$$\left| \frac{\sin x}{\sqrt{\cos x}} \right| < 2 \quad \forall x \in [0.5, 1]$$



The figure shows that this is true.

$\therefore$  FPI will converge for any initial point inside  $[0.5, 1]$

✓

4.2) You wish to find a root of the function

$$f(x) = x^3 - 2x + 1$$

using FPI with the auxiliary function

$$g(x) = \frac{x^3 + 1}{2}$$

- Prove that all of the fixed points of  $g(x)$  are roots of  $f(x)$ .
- Supposing we know that a fixed point of  $g$  exists in the interval  $[0, 0.7]$ , prove that FPI will converge for any initial condition in that interval.
- Create a table with the first two iterations of the fixed point iteration starting from  $x_0 = 0$

a)  $x$  is a f.p. of  $g(x) \Rightarrow \frac{x^3+1}{2} = x \Rightarrow x^3 - 2x + 1 = 0$   
 $\Rightarrow x$  is root of  $f(x)$  ✓

b)  $g'(x) = \frac{3}{2} x^2$

$\therefore \max_{[0, 0.7]} \frac{3}{2} x^2 = \frac{3}{2} (0.7)^2 = 0.735 < 1$  □

c)

| $k$ | $x_k$          | $g(x_k)$                                                              |
|-----|----------------|-----------------------------------------------------------------------|
| 0   | 0              | $\frac{1}{2}(0^3+1) = \frac{1}{2}$                                    |
| 1   | $\frac{1}{2}$  | $\frac{1}{2}\left(\left(\frac{1}{2}\right)^3+1\right) = \frac{9}{16}$ |
| 2   | $\frac{9}{16}$ | $\frac{1}{2}\left(\left(\frac{9}{16}\right)^3+1\right) = 0.589$       |

4.3) You wish to use a fixed-point iteration to find a root of the function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f(x_1, x_2) = \left( -x_1 + \frac{x_2}{2} + 1, x_1 x_2 - 1 \right)$$

You consider the auxiliary function  $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  (with  $x_1 \neq 0$ )

$$g(x_1, x_2) = \left( \frac{x_2}{2} + 1, \frac{1}{x_1} \right)$$

- a) Is this a valid choice, in the sense that the fixed points of  $g$  are roots of  $f$ ?
- b) Find a region of  $\mathbb{R}^2$  where the fixed point iteration is guaranteed to converge to a solution, provided one exists in that region.
- c) Pick an initial condition in the region you just found and carry out the first two iterations of FPI.

a)  $(x_1, x_2)$  is a f.p. of  $g \Rightarrow g(x_1, x_2) = (x_1, x_2)$

$$\Rightarrow \left( \frac{x_2}{2} + 1, \frac{1}{x_1} \right) = (x_1, x_2)$$
$$\Rightarrow \left( \frac{x_2}{2} + 1 - x_1, \frac{1}{x_1} - x_2 \right) = (0, 0)$$
$$\Rightarrow \left( \frac{x_2}{2} + 1 - x_1, 1 - x_1 x_2 \right) = (0, 0)$$
$$\Rightarrow \left( \frac{x_2}{2} + 1 - x_1, x_1 x_2 - 1 \right) = (0, 0)$$
$$\Rightarrow f(x_1, x_2) = (0, 0) \quad \checkmark$$

b) We want to find a region of  $\mathbb{R}^2$  where all partial derivatives have absolute value less than 1.

$$\bullet \frac{\partial g_1}{\partial x_1} = \frac{\partial}{\partial x_1} \left( \frac{x_2}{2} + 1 \right) = 0 \quad \dots \text{ok everywhere}$$

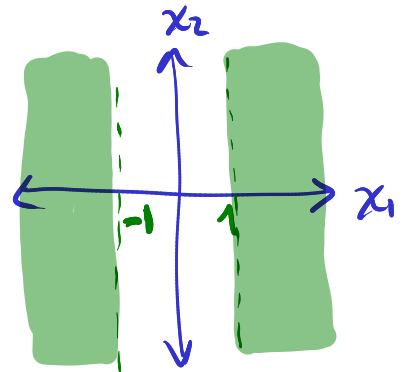
$$\bullet \frac{\partial g_1}{\partial x_2} = \frac{\partial}{\partial x_2} \left( \frac{x_2}{2} + 1 \right) = \frac{1}{2} \quad \dots \text{ok everywhere.}$$

- $\frac{\partial g_2}{\partial x_1} = \frac{\partial}{\partial x_1} \left( \frac{1}{x_1} \right) = -\frac{1}{x_1^2}$

$$\left| \frac{\partial g_1}{\partial x_1} \right| < 1 \iff \frac{1}{x_1^2} < 1 \iff x_1^2 > 1 \iff \begin{matrix} x_1 > 1 \\ \text{or} \\ x_1 < -1 \end{matrix}$$

- $\frac{\partial g_2}{\partial x_2} = \frac{\partial}{\partial x_2} \left( \frac{1}{x_1} \right) = 0 \dots$  ok everywhere.

$\therefore$  FPI will converge to a root of  $f$  whenever initialized with  $x_1 > 1$  or  $x_1 < -1$  provided a root of  $f$  exists in that region



4.4) Consider the function  $g(x) = 4x^3 + x^2$ .

a) What are the fixed points of  $g(x)$ ?

b) Create a table with the first three iterations of FPI with initial value  $x_0 = -0.5$ .

~~c) Find an interval for which it is guaranteed that if you start FPI in that interval, it will converge to a positive fixed point of  $g(x)$~~  **REMOVED**

a) Fixed points of  $g$  have  $g(x) = x$

$$4x^3 + x^2 = x$$

$$\Leftrightarrow x(4x^2 + x - 1) = 0.$$

$$\Leftrightarrow \begin{cases} x=0 \\ \text{or} \\ 4x^2 + x - 1 = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x=0 \\ \text{or} \\ x = \frac{-1 \pm \sqrt{17}}{8} \end{cases}$$

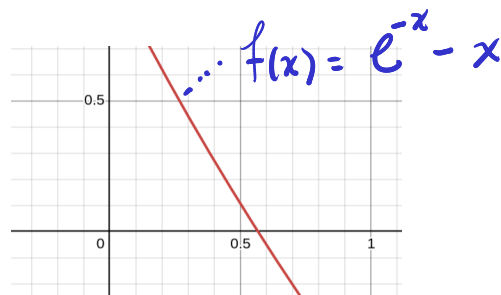
b)

| $k$ | $x_k$ | $g(x_k)$                           |
|-----|-------|------------------------------------|
| 0   | -0.5  | $4 \cdot (-0.5)^3 + 0.5^2 = -0.25$ |
| 1   | -0.25 | $4 \cdot (-0.25)^3 + 0.25^2 = 0$   |
| 2   | 0     | $4(0)^3 + 0^2 = 0$                 |

4.5) We wish to find a root of the function  $f(x) = e^{-x} - x$ , defined on the real line. Suppose that we know that a root of  $f$  exists in the interval  $[0.5, 1]$ . Prove that  $g(x) = e^{-x}$  is a good proxy function for finding this root using a fixed point iteration in the sense that a) a fixed point of  $g(x)$  is a root of  $f(x)$  and b) the fixed point iteration is guaranteed to converge.

$$f(x) = e^{-x} - x$$

$$g(x) = e^{-x}$$



a)

$$x \text{ is a fixed point of } g \Rightarrow g(x) = x$$

$$\Rightarrow e^{-x} = x$$

$$\Rightarrow e^{-x} - x = 0 \Rightarrow f(x) = 0$$

$$\Rightarrow x \text{ is root of } f.$$

b) We want to prove  $\max_{x \in [0.5, 1]} |g'(x)| < 1$

$$\text{With } g(x) = e^{-x} \text{ we have } |g'(x)| = e^{-x}$$

$$\therefore \max_{x \in [0.5, 1]} |g'(x)| = \max_{x \in [0.5, 1]} e^{-x}$$

$$= e^{-0.5} = 0.6 < 1$$





## 5) Week 13: Matrix operations

5.1) For the matrices  $A \in \mathbb{R}^{5 \times 2}$  and  $B \in \mathbb{R}^{2 \times 3}$ , is the matrix multiplication  $AB$  a valid operation? If so, what is the shape of  $AB$ ?

Yes, it is a valid operation.

$$\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \in \mathbb{R}^{5 \times 3}$$

$A \quad B$

5.2) What spaces do  $A$ ,  $x$ , and  $b$  belong to?

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}, \quad b = (8, 0), \quad x = \begin{bmatrix} 0 & 3 & 1 \end{bmatrix}$$

- ☐  $A \in \mathbb{R}^{3 \times 2}, b \in \mathbb{R}^2, x \in \mathbb{R}^3$
- ☐  $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^3, x \in \mathbb{R}^2$
- ☒  $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^2, x \in \mathbb{R}^3$
- ☐  $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^2, x \in \mathbb{R}^2$

One can also say  $x \in \mathbb{R}^{1 \times 3}$

5.3) Given the matrix  $A \in \mathbb{R}^{19 \times 23}$ , what space of matrices could  $B$  belong to in order for the product  $BA$  to be well-defined? Choose all that apply.

- ☒  $\mathbb{R}^{23 \times 19}$
- ☐  $\mathbb{R}^{23 \times 23}$
- ☐  $\mathbb{R}^{46 \times 23}$
- ☐  $\mathbb{R}^{19 \times 23}$

$$\begin{array}{cc} \xleftarrow{19} & \xleftarrow{23} \\ \left[ \begin{array}{c} B \in \mathbb{R}^{23 \times 19} \end{array} \right] & \left[ \begin{array}{c} A \in \mathbb{R}^{19 \times 23} \end{array} \right] \\ \uparrow & \uparrow \\ & 19 \end{array}$$

5.4) Given the following matrices  $G$  and  $H$ :

$$G = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, H = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$

- a) What is the value of  $HG$ ?
- b) What is the dimension of  $GH$ ?

$$a) \quad HG = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$= 2 \times 2 + 3 \times (-1) + 4 \times 3$$

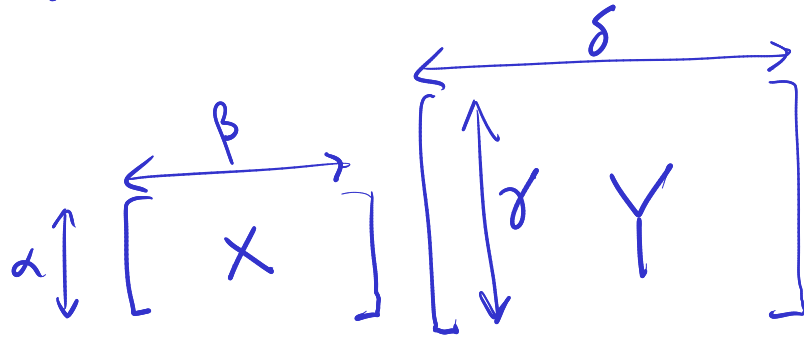
$$= 4 - 3 + 12 = 13$$

$$b) \quad \begin{matrix} G \in \mathbb{R}^{3 \times 1} \\ H \in \mathbb{R}^{1 \times 3} \end{matrix} \Rightarrow GH \in \mathbb{R}^{3 \times 3}$$

5.5) Given matrices  $A \in \mathbb{R}^{2 \times 4}$  and  $B \in \mathbb{R}^{\alpha \times \beta}$

- What should  $\alpha$  be so that  $AB$  is well-defined?
- What should  $\beta$  be so that  $BA$  is well-defined?

- In general, with  $X \in \mathbb{R}^{\alpha \times \beta}$  and  $Y \in \mathbb{R}^{\gamma \times \delta}$  we can multiply  $XY$  only if  $\beta = \gamma$ .
- That is, the number of columns of  $X$  must equal the number of rows in  $Y$ .



- With  $A \in \mathbb{R}^{2 \times 4}$ ,  $B \in \mathbb{R}^{\alpha \times \beta}$

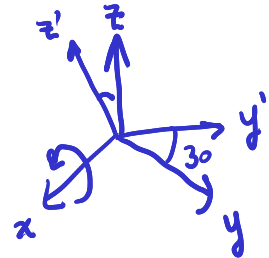
- $AB$  is defined only if  $\alpha = 4$
- $BA$  is defined only if  $\beta = 2$

## 6) Week 13: 3D Rotations and Projections

6.1) Let  $p = (1, 2, 1)$  represent the position of a penny in 3D space relative to the  $(x, y, z)$  coordinate frame. Find the position of the penny in the coordinate frame  $(x'', y'')$  obtained by applying the following transformations to  $(x, y, z)$ :

1. Rotate  $(x, y, z)$  by  $30^\circ$  about the  $x$  axis. This produces the  $(x', y', z')$  coordinate frame.
2. Perform a projection of the  $(x', y', z')$  coordinate frame onto  $(x'', y'')$  in which the  $z'$  coordinate becomes the  $y''$  coordinate ( $z' \rightarrow y''$ ), and the  $y'$  coordinate becomes the  $x''$  coordinate ( $y' \rightarrow x''$ ).

$$1. \quad \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30 & \sin 30 \\ 0 & -\sin 30 & \cos 30 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$



$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3}/2 & 1/2 \\ 0 & -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} + 1/2 \\ -1 + \sqrt{3}/2 \end{bmatrix}$$

$$2. \quad \begin{bmatrix} p_{x''} \\ p_{y''} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \sqrt{3} + 1/2 \\ -1 + \sqrt{3}/2 \end{bmatrix}$$

6.2) Let  $p = (p_x, p_y, p_z)$  represent a point in 3D space. Let

$$Z = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a matrix used to rotate the coordinate frame by  $\beta = 45^\circ$  around the  $z$  axis. Which of the following is the correct formula for determining  $p$ 's coordinates with respect to the rotated coordinate frame?

- ☐  $pZ$
- ☒  $Zp$
- ☐  $p + Z$

Note : This is following the convention of this class, set out in lecture, in which  $p = (p_x, p_y, p_z)$  should be arranged as a column vector and multiplied on the right side of the rotation matrix  $Z$ .

If you have learned a different convention in another class, you must still do it this way on the final.

## 7) Week 13: Systems of linear equations

7.1) Express the following system of equations in matrix form.

$$1 - x_2 = 3x_3$$

$$x_1 = 7 - x_2$$

$$x_1 + x_2 + x_3 = 0$$

Organize variables on the left and constants on the right.

$$\begin{cases} -x_2 - 3x_3 = -1 \\ x_1 + x_2 = 7 \\ x_1 + x_2 + x_3 = 0 \end{cases}$$

Then with  $X = (x_1, x_2, x_3)$

$$\begin{bmatrix} 0 & -1 & -3 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \\ 0 \end{bmatrix}$$

$\vdots$                        $\vdots$                        $\vdots$   
 $A$                        $X$                        $b$

7.2) A system of linear equations  $Ax = b$  is known to have infinitely many solutions. What can be said about the rank of  $A$ ?

The rank of  $A$  must be less than  $n$ , the number of unknowns.

7.3) Express the following system of equations in matrix form.

$$x_1 + x_2 = 1$$

$$x_1 + x_3 = 2$$

$$x_2 + x_3 = 3$$

$$2x_1 + 2x_2 + 2x_3 = 4$$

With  $X = (x_1, x_2, x_3)$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

7.4) A system of linear equations is given in matrix form  $Ax = b$ , with  $A \in \mathbb{R}^{2 \times 5}$ .

- a) What is the dimension of  $x$ ?
- b) What is the dimension of  $b$ ?
- c) How many solutions are there to the system of equations if both  $\text{rank}(A)$  and  $\text{rank}(A|b)$  equal 2?

$$\begin{bmatrix} \vdots & \vdots & A & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \vdots \\ x \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ b \\ \vdots \end{bmatrix}$$

a)  $x \in \mathbb{R}^5 \quad \dots \quad n=5$

b)  $b \in \mathbb{R}^2$

c)  $\text{rank}(A) = \text{rank}(A|b) = 2 < n$ .

$\Rightarrow$  infinitely many solutions.

7.5) The rank of  $A$  is 3 and the rank of  $A|b$  is 4. How many solutions are there to this system of equations  $Ax = b$ ?

$$\begin{aligned}\text{rank}(A) &= 3 \\ \text{rank}(A|b) &= 4\end{aligned}$$

$$\text{rank}(A|b) > \text{rank}(A) \Rightarrow 0 \text{ solutions.}$$

7.6) You are given the matrix equation  $Ax = b$ , where  $b$  is not in the span of  $A$ . How many possible  $x$  satisfy this equation?

$$\text{Since } b \notin \text{span}(A), \text{ we have } \text{rank}(A|b) > \text{rank}(A)$$

$$\Rightarrow \text{no solutions.}$$

.



## 8) Week 14: Initial value problems

**8.1)** Consider the following ordinary differential equation:

$$y'' + \alpha y'y + y = t$$

- With  $\alpha = 1$ , is the ODE *linear* or *nonlinear*?
- With  $\alpha = 0$ , is the ODE *linear* or *nonlinear*?
- With  $\alpha = 1$ , what is the order of the ODE?
- With  $\alpha = 0$ , what is the order of the ODE?
- Express the ODE in *vector form* (keeping  $\alpha$  as a variable).
- Use Euler's method to estimate the value of  $y(1)$ , starting from  $y(0) = 2$ ,  $y'(0) = 1$ , and with a step size of  $h = 1$ . Express the solution as a function of  $\alpha$ .

a)  $\alpha=1$  :  $y'' + \underbrace{y'y}_{\text{nonlinearity}} + y = t$  ... nonlinear

b)  $\alpha=0$   $y'' + y = t$  ... linear.

c) 2nd order

d) 2nd order

e)  $y_1 = y$  ,  $y_2 = y'$  ,  $\gamma = (y_1, y_2)$

$$\begin{aligned} \therefore \gamma' &= (y'_1, y'_2) = (y_2, y'') \\ &= (y_2, -\alpha y'_1 y_1 - y_1 + t) \\ &= (y_2, -\alpha y_2 y_1 - y_1 + t) = f(\gamma, t). \end{aligned}$$

or equivalently:

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ -\alpha y_2 y_1 - y_1 + t \end{bmatrix}$$

$$f) \quad \hat{Y}[0] = \begin{bmatrix} y'(0) \\ -\alpha y'(0) y(0) - y(0) + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -\alpha \cdot 1 \cdot 2 - 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2\alpha - 2 \end{bmatrix}$$

$$\hat{Y}[1] = \hat{Y}[0] + f(\hat{Y}[0], 0) \cdot h$$

$$= \begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix} + f\left(\begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix}, 0\right) \cdot (1)$$

$$= \begin{bmatrix} \hat{y}_1[0] \\ \hat{y}_2[0] \end{bmatrix} + \begin{bmatrix} \hat{y}_2[0] \\ -\alpha \hat{y}_2[0] \hat{y}_1[0] - \hat{y}_1[0] + 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -\alpha(1)(2) - 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2\alpha - 1 \end{bmatrix}$$

$$\therefore \hat{y}(1) = 3$$

8.2) Rewrite the following second-order linear ODE into a system of first-order ODEs.

$$y'' - y' + y = 0$$

- a) Is this a linear ODE or a non-linear ODE?
- b) Write the ODE in matrix form.
- c) With initial condition  $y(0) = 1$ ,  $y'(0) = 1$  and step size  $h = 1$ , perform one step of Euler's method and estimate  $y(1)$ .
- d) Perform one step of the midpoint method.
- e) Perform one step of Heun's method.

a) linear

b)  $\gamma = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$  with  $y_1 = y$ ,  $y_2 = y'$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ y_2 - y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\therefore \gamma' = A\gamma \text{ with } A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

c) Euler:

$$\hat{\gamma}_{[0]} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \hat{\gamma}_{[1]} &= \hat{\gamma}_{[0]} + h f(\hat{\gamma}_{[0]}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

d) Midpoint

$$S_0 = f(\hat{\gamma}_{[0]}) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\gamma_m = \hat{\gamma}_{[0]} + \frac{h}{2} S_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$$

$$S_m = f(y_m) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \hat{S}$$

$$\hat{y}_{[1]} = \hat{y}_{[0]} + h \hat{S} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

e) Heun.

$$S_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$y_1 = \hat{y}_{[0]} + h S_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$S_1 = f(y_1) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\hat{S} = \frac{1}{2} (S_0 + S_1) = \frac{1}{2} \left( \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}.$$

$$\hat{y}_{[1]} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$$

8.3) Consider the following ODE:

$$y''' + 3t^2 y' - 2y = e^{-t}$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the Heun's method with initial condition  $y = 0$ ,  $y' = 1$ ,  $y'' = 0$ , and time step  $h = 1$ .

a) Third order.

b) Linear

c)  $y_1 = y$        $Y = (y_1, y_2, y_3)$   
 $y_2 = y'$   
 $y_3 = y''$

$$\begin{aligned} Y' = (y_1', y_2', y_3') &= (y_2, y_3, e^{-t} - 3t^2 y' + 2y) \\ &= (y_2, y_3, e^{-t} - 3t^2 y_2 + 2y_1) \end{aligned}$$

Matrix form:

$$\underbrace{\begin{bmatrix} y_1' \\ y_2' \\ y_3' \end{bmatrix}}_{Y'} = \begin{bmatrix} y_2 \\ y_3 \\ e^{-t} - 3t^2 y_2 + 2y_1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -3t^2 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_Y + \underbrace{\begin{bmatrix} 0 \\ 0 \\ e^{-t} \end{bmatrix}}_W$$

$$f(Y, t) = A(t) Y + w(t)$$

$$d) \quad \hat{y}[0] = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} s_0 &= f(\hat{y}[0], 0) = A(0) \hat{y}[0] + W(0) \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$y_1 = \hat{y}[0] + h s_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} s_1 &= f(y_1, h) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -3 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ e^{-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/e \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 + 1/e \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \hat{s} &= \frac{1}{2} (s_0 + s_1) = \frac{1}{2} \left( \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -1 + 1/e \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 \\ 1/2 \\ 1/2e \end{bmatrix} \end{aligned}$$

$$\hat{y}[1] = \hat{y}[0] + h \hat{s} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1/2 \\ 1/2e \end{bmatrix} = \begin{bmatrix} 2 \\ 1/2 \\ 1 + 1/2e \end{bmatrix}$$

8.4) The Duffing equation is used to model certain damped and driven oscillators. The Duffing equation is given by:

$$\beta y^3 + y'' + \delta y' + \alpha y = \gamma \cos(\omega t)$$

where  $\delta$  is the damping coefficient,  $\alpha$  and  $\beta$  are the linear and non-linear stiffness coefficients, respectively, and  $\gamma \cos(\omega t)$  represents the driving force with amplitude  $\gamma$  and frequency  $\omega$ .

- What is its order?
- Is it linear or nonlinear?
- Express the ODE in vector form. If linear, express it in matrix form.

a) Second order

b) Nonlinear.

$$\begin{aligned} \text{c) } \begin{matrix} y_1 = y \\ y_2 = y' \end{matrix} &\longrightarrow \begin{matrix} y_1' = y_2 \\ y_2' = y'' = \gamma \cos \omega t - \alpha y - \delta y' - \beta y^3 \\ &= \gamma \cos \omega t - \alpha y_1 - \delta y_2 - \beta y_1^3 \end{matrix} \end{aligned}$$

$$Y = (y_1, y_2)$$

$$Y' = (y_1', y_2') = (y_2, \gamma \cos \omega t - \alpha y_1 - \delta y_2 - \beta y_1^3)$$

8.5) Consider the following ODE:

$$y'' - \mu(1 - y^2)y' + y = 0$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the midpoint method with initial condition  $y = 0$ ,  $y' = 1$ , time step  $h = 0.5$ , and  $\mu = 1$ .

a) 2nd order.

b) non linear.

c)  $y_1 = y$   
 $y_2 = y'$        $\gamma = (y_1, y_2)$

$$\gamma' = (y_1', y_2') = (y_2, \mu(1 - y_1^2)y_2 - y_1) = f(\gamma)$$

d)  $f((y_1, y_2)) = (y_2, (1 - y_1^2)y_2 - y_1)$

$$\hat{\gamma}[0] = (0, 1)$$

$$S_0 = f(\hat{\gamma}[0]) = f((0, 1)) = (1, (1 - 0)1 - 0) \\ = (1, 1)$$

$$\gamma_m = \hat{\gamma}[0] + h S_0 = (0, 1) + \frac{0.5}{2} (1, 1) \\ = (0.25, 1.25)$$



$$\begin{aligned}
 S_m = f(y_m) &= f((0.25, 1.25)) \\
 &= (1.25, (1 - 0.25^2)(1.25) - 0.25) \\
 &= (1.25, 0.921875) = \hat{S}
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}[1] &= \hat{y}[0] + 0.5 \hat{S} \\
 &= (0, 1) + 0.5 (1.25, 0.921875) \\
 &= (0.625, 1.461)
 \end{aligned}$$

8.6) Consider the following ODE:

$$y'' - 4y' + 4y = 10$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first two iterations of Euler's method with initial condition  $y = 0$ ,  $y' = 0$  and time step  $h = 0.1$ .

a) 2nd order

b) linear

c) 
$$\begin{aligned} y_1 &= y \\ y_2 &= y' \end{aligned} \rightarrow Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$Y' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ 10 + 4y_2 - 4y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} Y + \begin{bmatrix} 0 \\ 10 \end{bmatrix}$$

d) 
$$\hat{Y}[0] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \hat{Y}[1] &= \hat{Y}[0] + h f(\hat{Y}[0]) \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0.1 \left( \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} . \end{aligned}$$

$$\begin{aligned} \hat{Y}[2] &= \hat{Y}[1] + h f(\hat{Y}[1]) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.1 \left( \begin{bmatrix} 0 & 1 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0.1 \left( \begin{bmatrix} 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 1.4 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 2.4 \end{bmatrix} \end{aligned}$$

## 9) Week 15: Randomness

9.1) The variable  $X$  is uniformly distributed over the interval  $[1, 3]$ .

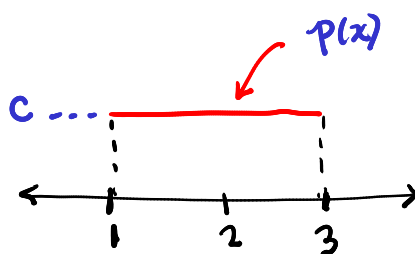
- Give a formula for the distribution of  $X$ .
- Compute the probability of the event  $e = [\frac{3}{2}, \frac{5}{2}]$ .

a)

• Random variable  $X$  has sample space  $\Omega = [1, 3]$

• Uniform distribution:

$$c = \frac{1}{b-a} = \frac{1}{3-1} = \frac{1}{2}$$

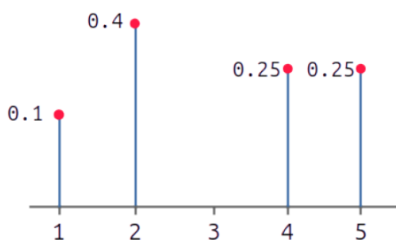


$$p(x) : [1, 3] \rightarrow \mathbb{R}$$
$$x \mapsto \frac{1}{2}$$

$$\begin{aligned} \text{b) } P\left(\left[\frac{3}{2}, \frac{5}{2}\right]\right) &= \int_{3/2}^{5/2} p(x) dx \\ &= \int_{3/2}^{5/2} \frac{1}{2} dx \\ &= \frac{1}{2} \end{aligned}$$



9.2) For the discrete distribution shown in the figure below, answer the following.



- What is its sample space?
- Calculate its mean.
- Calculate its standard deviation.
- What is the probability of the event  $\{1, 2\}$ ?

$$a) \Omega = \{1, 2, 4, 5\}$$

$$b) \mu_x = \sum_{x \in \Omega} x p(x)$$

$$= 1(0.1) + 2(0.4) + 4(0.25) + 5(0.25) = 3.15$$

$$c) \sigma_x = \sqrt{E[(X - \mu_x)^2]}$$

Let's make a table for the r.v.  $(X - \mu_x)^2$

| $x$ | $p(x)$ | $(x - \mu_x)^2$       |
|-----|--------|-----------------------|
| 1   | 0.1    | $(1 - 3.15)^2 = 4.62$ |
| 2   | 0.4    | $(2 - 3.15)^2 = 1.32$ |
| 4   | 0.25   | $(4 - 3.15)^2 = 0.72$ |
| 5   | 0.25   | $(5 - 3.15)^2 = 3.42$ |

$$\sigma_x^2 = E[(X - \mu_x)^2] = 4.62(0.1) + 1.32(0.4) + 0.72(0.25) + 3.42(0.25) = 2.03$$

$$\therefore \sigma_x = 1.42$$

9.3) The function  $p(x) = c x^2$  is defined on the interval  $\Omega = [0, 3]$ .

- a) What must the value of  $c$  be for  $p(x)$  to be a valid distribution?
- b) Find the mean of  $X$  (the random variable whose distribution is  $p(x)$ ).
- c) What is the probability of sampling a value of  $X$  in the interval  $[1, 2]$ ?

a) Positivity:  $p(x) \geq 0 \quad \forall x \in \Omega$

$$\Leftrightarrow c x^2 \geq 0 \quad \forall x \in [0, 3]$$

$$\Leftrightarrow c \geq 0$$

Unit integral  $\int_{\Omega} p(x) = 1$

$$\Leftrightarrow \int_0^3 c x^2 = 1$$

$$\Leftrightarrow c \left. \frac{x^3}{3} \right|_0^3 = 1$$

$$\Leftrightarrow c \cdot 9 = 1 \quad \Leftrightarrow c = \frac{1}{9} \quad \checkmark$$

b)  $\mu_x = \int_0^3 x p(x) dx$

$$= \int_0^3 \frac{1}{9} x^3 dx = \frac{1}{9} \left. \frac{x^4}{4} \right|_0^3 = \frac{81}{36} = \frac{9}{4}$$

c)  $P([1, 2]) = \int_1^2 p(x) dx = \int_1^2 \frac{1}{9} x^2 dx = \frac{1}{9} \left. \frac{x^3}{3} \right|_1^2$

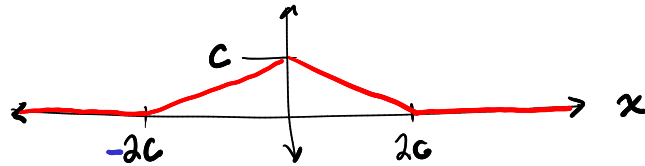
$$= \frac{8-1}{27} = \frac{7}{27}$$

9.4) The function  $p(x)$  is defined on the real line.

$$p(x) = \max\left(0, c - \frac{|x|}{2}\right)$$

- Draw this function for  $c = 1$ .
- What must  $c$  be for  $p(x)$  to be a valid distribution?
- With  $c$  set as in part b), what is the probability of the event  $[-c, c]$ ?

a)



b) Positivity:  $c \geq 0$

Unit integral:  $\int_{-\infty}^{\infty} p(x) dx = 1$

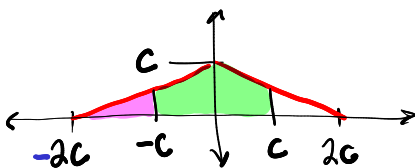
$\Leftrightarrow$  triangular area  $= 1$

$\Leftrightarrow \frac{1}{2} \cdot 4c \cdot c = 1$

$\Leftrightarrow 2c^2 = 1 \Leftrightarrow c = \pm \sqrt{1/2} \dots$

Combining both  
we have  
 $c = \sqrt{1/2}$

c)  $P([-c, c]) = \int_{-c}^c p(x) dx = \text{green area}$



$= 1 - 2 \times (\text{pink area})$

$= 1 - 2 \left( \frac{1}{2} \cdot c \cdot \frac{c}{2} \right) = 1 - \frac{c^2}{2}$

$= 1 - \frac{1}{4} = \frac{3}{4}$

9.5) A Monte Carlo simulation has been run to estimate the variation of the temperature at the center of a heated plate when the temperatures at its corners are subjected to random variations. Upon running 3 executions of the code, we obtain the following dataset (expressed in degrees Celsius).

$$\mathcal{D} = \{51.7, 53.6, 52.9\}$$

- a) Compute the sample mean.
- b) Compute the sample standard deviation.
- c) Estimate the probability that the temperature falls between 51°C and 53°C.

$$n = 3$$

$$a) \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{3} (51.7 + 53.6 + 52.9) = 52.7$$

$$b) \hat{\sigma}_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_n)^2}$$

$$= \sqrt{\frac{1}{2} \left( (51.7 - 52.7)^2 + (53.6 - 52.7)^2 + (52.9 - 52.7)^2 \right)}$$

$$= 0.96$$

$$c) e = [51, 53]$$

$$\hat{p}(e) = \frac{\# \text{ samples in } e}{n} = \frac{2}{3}$$